

Educore Enterprise AI Framework

System Deployment Architectures & Resource Sizing Analysis

Strategic Engineering & Sizing Guide for 5,000 Registered Users & up to 2,500 Concurrent Sessions

ISO/IEC 42001:2023 AIMS

ZAMBIAN DPA NO. 3 OF 2021

EU AI ACT (ART. 5)

EDU-AIMS-HBK-V1.0 & POL-V1.0

MULTI-CAMPUS RBAC

5,000

TOTAL USER DIRECTORY

2,500

PEAK CONCURRENT SESSIONS

100 QPS

SUSTAINED QUERY LOAD

400–800

SIMULTANEOUS LLM STREAMS

1. Grounding in the Actual Framework (EDUCORE_AI_FRAMEWORK)

This technical analysis directly references the official governance corpus maintained in `EDUCORE_AI_FRAMEWORK`, consisting of the Master Governing Handbook (`EDU-AIMS-HBK-v1.0/v2.0`), the Central Organizational Policy (`EDU-AIMS-POL-v1.0/v3.0`), the 180-Day Implementation Action Plan (`EDU-AIMS-PLN`), and the Academic Cognitive Bypass Audit Checklist (`EDU-AIMS-CHK`).

1.1 Dual-Track Organizational Charter

The Educore AI Management System (AIMS) governs two strictly differentiated institutional tracks:

- Track 1: Enterprise Business Operations** — Governs Finance, Procurement, IT/DevOps, Human Resources, Legal, Facilities, and Central Administration. Focuses on data loss prevention, financial verification, secret scanning, and Zambian statutory data residency.
- Track 2: Academic & Pedagogical Transformation** — Governs Teaching Faculty, Curriculum Heads, Teacher Academies, and Students across Sentinel Kabitaka (SKAB S, SKAB P), Sentinel Kalumbila (SKAL), Trident College (TCL), Trident Preps (TPS, TPK, TPL), and Frontier Nkisu. Focuses on Cambridge 0580 syllabus alignment, human-in-the-loop review, Adelaide-model intellectual ownership declarations, and Viva Voce oral defenses.

1.2 Microsoft Purview Data Containers & Clearances

The framework enforces strict Zero-Trust boundaries mapped across four sensitivity containers:

PURVIEW SENSITIVITY LABEL	CLEARANCE TIER	PERMITTED ROLES & PERSONAS	DATA SCOPE & CONTENT BOUNDARIES
Public / Educational	Tier C (Public)	Students, External Parents, Public	Published syllabi, textbook indices, public announcements, and Socratic diagnostic tutors.
Internal - Educational	Tier B (Staff)	Teaching Faculty, Curriculum Leads	Curriculum maps, lesson designs, assessment rubrics, and internal staff notices.
Confidential - Admin / Finance	Tier A (Confidential)	Bursars, HR, Pastoral, Legal, Execs	Financial variance ledgers, student fee arrears, payroll, bursaries, and safeguarding records.
Restricted - IT / Systems	Tier A (Restricted IT)	IT Engineers, Systems Administrators	Network topologies, server configs, API keys, database connection strings, and security repos.

1.3 8 Departmental Non-Software Guardrails

The backend enforces procedural stop-conditions that software firewalls cannot intercept:

- **Fin-01 (Dual-Key Manual Audit)** : Independent human calculation audit notice appended to all AI-generated financial amounts.
- **Fin-02 (Zambian DPIA Compliance)** : Regex redaction of Zambian National Registration Card (NRC) numbers (`\b\d{6}/\d{2}/\d{1}\b`) and FQM mining partner subsidies.
- **IT-01 (Pre-Commit Secret Scanning)** : Automated detection and blocking of AWS credentials, RSA private keys, and database connection URIs in user prompts.
- **HR-01 (Algorithmic Autonomy)** : Absolute prohibition on automated candidate shortlisting, rejection, or appraisal scoring.
- **Edu-01 & Edu-02 (Curriculum Verification)** : Prohibition of automated summative exam grading; mandatory alignment against Cambridge 0580 syllabi.
- **Stu-01 (Cognitive Bypass Prevention)** : Diagnostic hints only for students; homework answer dumping is strictly blocked.
- **Fac-01 & Fac-02 (Physical Safety)** : Ban on campus gate facial recognition/emotion tracking; instant human emergency dispatch for hazards.
- **Leg-01 & Leg-02 (Legal Accountability)** : Statutory verification against Zambian law books; non-delegable executive accountability.

2. Prototype Limitations vs. Production Concurrency

The current repository implementation in `src/backend/educore_enterprise_backend.py` serves as an operational proof-of-concept. To support **2,500 concurrent active users** and **5,000 registered accounts**, several core bottlenecks must be transitioned:

SUBSYSTEM	CURRENT PROTOTYPE STATE	FAILURE MODE UNDER 2,500 CONCURRENT USERS	TARGET PRODUCTION REPLACEMENT
Web Listener	Python <code>ThreadingHTTPServer</code>	Thread-per-connection model exhausts OS threads and file descriptors beyond ~300 concurrent SSE streams.	Asynchronous ASGI (FastAPI + Uvicorn workers) behind Traefik/Nginx with HTTP/2 multiplexing.
Vector Retrieval	ChromaDB with global <code>threading.RLock()</code>	All similarity searches are serialized; 100 QPS creates massive thread contention and latency spikes.	Distributed vector cluster (Qdrant or PostgreSQL <code>pgvector</code>) with in-memory HNSW index.
LLM Inference	Local CPU Ollama (<code>llama3.2:1b</code> , 6 threads)	CPU processes 1 generation stream sequentially (~20 tok/s). 2,500 users experience multi-hour queues.	Continuous-batching GPU cluster (vLLM / TensorRT-LLM) or Azure OpenAI Provisioned Throughput.
State Database	Single SQLite file (<code>webui.db</code>)	SQLite locks on concurrent user writes, causing <code>database is locked</code> crashes.	High-availability PostgreSQL cluster with PgBouncer connection pooling.
Audit Ledger	Append-only flat file (<code>aims_rag_audit.jsonl</code>)	File lock contention and disk I/O bottlenecks at 100+ audit writes per second.	Clustered PostgreSQL or ClickHouse with immutable write-ahead logging (ISO 42001 Clause 7.5).

3. Three Production Deployment Paths

Path A: Enterprise Hybrid Cloud Azure OpenAI + Azure AI Search + Entra ID	Path B: Sovereign Private Cluster On-Premises K8s + vLLM GPU Nodes	Path C: Campus Hub-and-Spoke Central Admin Hub + Edge Campus Appliances
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Path A: Enterprise Hybrid Cloud (Framework Reference Architecture)

Implements the Microsoft Cloud topology codified in the Master Implementation Handbook (Section 2):

- **Identity & Access:** Microsoft Entra ID (Azure AD) SSO integration. User groups (Students, Faculty, Bursary, IT, Admin) map directly to Educore clearance tiers.
- **Frontend:** Open WebUI deployed on **Azure Container Apps** with autoscaling (4 to 10 replicas) behind Azure Front Door.
- **Governance Layer:** Stateless FastAPI middleware executing the 8 guardrails, Zambian NRC masking, and Purview container gating.
- **Retrieval & Purview:** **Azure AI Search** (Standard S2 tier) indexing SharePoint Online libraries, natively enforcing Purview sensitivity labels.

- **Inference: Azure OpenAI Service** deploying `gpt-4o-mini` (for universal RAG & Socratic tutoring) and `gpt-4o` (for executive/finance tasks) under Microsoft Enterprise Data Protection (zero customer data retention).

Path B: Sovereign On-Premises Private Cloud Cluster

Industrializes the current Python/Chroma/Open WebUI codebase into a sovereign Kubernetes cluster hosted in an Educare data center or Zambian tier-3 colocation facility:

- **Orchestration & Ingress:** Kubernetes (k3s / RKE2) with Traefik Ingress, SSL termination, and rate limiting.
- **Frontend & Gateway:** 6x Open WebUI replicas + 8x Educare Governance Gateway worker pods with Redis session caching.
- **Vector Database:** 3-node **Qdrant Distributed** or **PostgreSQL with pgvector** cluster with HNSW in-memory indexing.
- **Inference Serving:** 2x 8-GPU inference nodes running **vLLM** with PagedAttention and continuous batching over Llama 3.1 8B Instruct (FP8).
- **Audit Store:** Clustered PostgreSQL or ClickHouse with immutable write-ahead logging.

Path C: Distributed Multi-Campus Hub-and-Spoke Architecture

Reflects Educare's geographical spread across Zambia (Sentinel Kabitaka, Sentinel Kalumbila, Trident College, Trident Preps, Frontier Nkisu):

- **Central Hub (Central Administration):** Houses master corpus synchronization (`FrameworkSyncService`), global ISO 42001 audit ledger, Entra ID master directory, and high-tier executive models.
- **Campus Spokes (Local Edge Servers):** Edge appliances (e.g. 2U servers with dual GPUs) deployed at each campus site. Serves local Open WebUI, local Socratic student tutoring, and local cached vectors with zero WAN latency.

4. Resource Requirements: 5,000 Users / 2,500 Concurrent

4.1 Concurrency & Workload Mathematical Modeling

In an educational and enterprise environment, 2,500 concurrent active users with open browser sessions exhibit the following load dynamics:

- **Human Query Cadence:** Users type, read, and reflect, querying every **25 seconds** on average.
- **Sustained Request Rate:** $2,500 \text{ users} / 25 \text{ seconds} = 100 \text{ Queries per Second (QPS)}$.
- **Peak Burst Factor (2.0x):** **200 QPS** during class start times or synchronized exam prep.
- **Simultaneous Active LLM Streams (Little's Law, $L = \lambda W$):** With an average generation duration of 4 seconds (~150–200 tokens):
 $L(\text{sustained}) = 100 \text{ QPS} * 4s = 400 \text{ simultaneous generating streams}$
 $L(\text{peak}) = 200 \text{ QPS} * 4s = 800 \text{ simultaneous generating streams}$

4.2 Hardware & Infrastructure Specifications

On-Premises Private Cloud Cluster (Path B)

ARCHITECTURE LAYER	NODE / INSTANCE COUNT	COMPUTE & HARDWARE SPECIFICATIONS	OPERATIONAL ROLE & CAPACITY
Ingress & Load Balancer	2 Nodes (Active/Passive HA)	8 vCPU, 16 GB RAM, 10 GbE NIC	SSL termination, rate limiting, sticky SSE streaming connections.
Open WebUI Frontend	6 Container Replicas	4 vCPU, 8 GB RAM per replica	Handles web sessions, model dropdowns, and chat history rendering.
Educore Governance Gateway	8 Worker Pods (FastAPI)	4 vCPU, 8 GB RAM per pod	Stateless enforcement of 8 Guardrails, NRC masking, Purview RBAC gating.
Vector Search (Qdrant)	3 Nodes (Clustered)	8 vCPU, 32 GB RAM, NVMe SSD	HNSW in-memory dense vector search; sub-10ms retrieval at 100 QPS.
Embedding Inference	2 Dedicated GPUs (or 32 vCPU)	2x NVIDIA L4 (24 GB)	Real-time query embedding generation via nomic-embed-text at 100 QPS.
LLM Inference (vLLM Cluster)	2x 8-GPU Servers (16 GPUs Total)	16x NVIDIA L40S (48 GB) or 8x A100 (80 GB) 2x Dual Xeon/EPYC, 512 GB RAM	Continuous batching for 400–800 concurrent streams; Llama 3.1 8B FP8; 768 GB total VRAM for model weights and 2,048-token KV cache.
Session & State (PostgreSQL/Redis)	2 Nodes (Primary + Replica)	PostgreSQL: 8 vCPU, 32 GB RAM, NVMe Redis: 16 GB RAM Cluster	Redis handles user session tokens and SSE streaming state; PostgreSQL stores user groups, ACLs, and chat histories.
ISO 42001 Audit Ledger	2 Nodes (ClickHouse)	8 vCPU, 32 GB RAM, 1 TB NVMe	100–200 audit writes/sec with real-time compliance querying.

Enterprise Hybrid Cloud (Path A - Azure)

AZURE SERVICE	TARGET SKU / PROVISIONING	ROLE IN ARCHITECTURE
Azure Front Door	Standard Tier	Global WAF, SSL offloading, DDoS protection, edge caching.
Azure Container Apps (WebUI)	4 to 10 Replicas (2 vCPU, 4 GB RAM)	Autoscaling Open WebUI instances with Entra ID SSO integration.
Azure Container Apps (Gateway)	4 to 12 Replicas (2 vCPU, 4 GB RAM)	Stateless Educore Governance Middleware with async connection pooling.
Azure AI Search	Standard S2 (3 Replicas / 2 Partitions)	Purview-labeled RAG retrieval handling 100+ QPS with semantic ranker.
Azure OpenAI Service	200–300 Provisioned Throughput Units (PTUs)	Guaranteed throughput for gpt-4o-mini; ~35,000 tokens/second at peak.
Azure Database for PostgreSQL	Flexible Server (8 vCPU, 32 GB RAM)	Shared database for Open WebUI user directory and ISO 42001 audit logs.
Azure Cache for Redis	Premium P1 (6 GB)	Session state caching, rate limiting, and prompt caching.

4.3 Network & Bandwidth Sizing

- **Streaming Egress Bandwidth:** 400 active generation streams × 20 tokens/sec × 4 bytes/token ≈ 32 KB/sec raw payload. Including HTTP/2 SSE headers, TCP overhead, and WebUI polling: **~15 to 30 Mbps sustained egress.**
- **Campus LAN:** 1 Gbps switched backbone between access points and local gateways.
- **Campus WAN Uplink (Path A):** Minimum **100 Mbps symmetrical dedicated fiber** per campus with QoS prioritizing AI API endpoints.

5. Implementation Roadmap (180-Day Action Plan)

Aligned with the 5-phase sequence established in `EDU-AIMS-PLN` :

PHASE & TIMELINE	MILESTONES & DELIVERABLES	ISO 42001 & STATUTORY GOVERNANCE ALIGNMENT
Phase 1 (Days 1–30) Governance Scaffolding	Refactor <code>educore_enterprise_backend.py</code> to async ASGI (FastAPI). Integrate Entra ID SSO. Deploy CASB/DNS discovery and run the 10-Question Discovery Survey across departments.	ISO 42001 Clause 4.3; Asset Inventory; Zambian NRC regex masking verification.
Phase 2 (Days 31–60) Storage & Scale-Out	Decouple local ChromaDB; migrate to distributed Qdrant or Azure AI Search. Ingest Microsoft Purview labels across SharePoint. Migrate SQLite to PostgreSQL.	ISO 42001 Clause 6.1 (Risk Treatment); Statutory Zambian DPIA under Act No. 3 of 2021.
Phase 3 (Days 61–90) Inference Deployment & Pilots	Commission vLLM GPU cluster or Azure OpenAI PTUs. Launch Tier A (Admin/Finance) and Tier B (Faculty) pilots. Run Michigan-style AI literacy workshops.	ISO 42001 Clause 7.2 (Competence); Fin-01 dual-key validation; Edu-02 syllabus check.
Phase 4 (Days 91–120) Student Socratic Rollout	Enable <code>educore-socratic-student</code> for Tier C learners across Sentinel, Trident, and Frontier. Enforce Adelaide-model ownership pledges and Viva Voce oral spot-checks.	Guardrail Stu-01; 4-Vector Cognitive Bypass Audits (EDU-AIMS-CHK); Scale to 2,500 concurrency.
Phase 5 (Days 121–180) AIMS Audit & Certification	Execute comprehensive ISO 42001 internal audit across Clauses 4–10. Convene CEO Top-Management Review. Undergo Stage 1/2 certification audits with accredited body.	ISO/IEC 42001:2023 Official Enterprise Certification; Ongoing Purview Event Logging.

IMMEDIATE TECHNICAL NEXT STEPS

- Asynchronous ASGI Migration:** Replace `http.server.ThreadingHTTPServer` in `src/backend/educore_enterprise_backend.py` with FastAPI/Uvicorn to unlock non-blocking I/O for 2,500 SSE connections.
- Vector Store Externalization:** Eliminate `_CHROMA_LOCK` by pointing retrieval calls to an external Qdrant or Azure AI Search cluster.
- Database Migration:** Direct Open WebUI and backend user/session data from local SQLite to PostgreSQL with connection pooling.